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ASSIGNMENT 2

Excavator (Hydraulic Equipment)

Excavators are earthmoving heavy construction equipment consisting of a boom, dipper, bucket and cab on a rotating platform known as the "house". The house sits on top of the undercarriage with tracks and wheels. They are natural progression from the steam shovels.

The operator of the excavator uses pedals and levers to steer the equipment, and to move forward and backwards. Excavator are activated by the diesel-powered engine, and hydraulic motors control the arm of the excavator.

The three main sections of an excavator are the undercarriage, the house and the arm. The undercarriage is the lowest section responsible for moving the excavator around the area. The house sits above the undercarriage and contains the power source and operator controls. The arm projects from the main body of the excavator and is responsible for performing most actual physical work.

The excavator arm is incorporated into the lower frame chassis. It is composed of three hydraulic pistons that operate in tandem with steel piston arms.

The two main sections and bucket loader are connected by means of a hinge. The extension is the first piston widens the section at the same time as increases in girth, providing further reach for the operator. The third piston activates the bucket loader, enabling the collection of debris into the bucket.

The excavator attachments require fixed tracks in order to function. The tracks are operated by means of manipulating gears that are fused with the engine. Use of the gears allows the hydraulic excavator to move backwards and forwards. Turning the machine involves replacing one track with another. More complex pieces of equipment involve manipulating more than one track at the same time.

The hydraulic pumps squeeze the fluid in the hydraulic system through small diameter hoses and pipes, this increases the pressure of fluid to create mechanical force. The force of the fluid at the working end of the lines activates pistons or other moveable surfaces, and operation begin to occur ,thus, the bucket is drawn through the soil, a thumb clamp down etc.

Jack Hammer (Pneumatic Equipment)

Jack Hammer is a pneumatic tool that combines a hammer directly with a chisel. The hand held jackhammer is powered by compressed air. The jackhammer is used to break-up rock, pavement and concrete.

The jackhammer operate by driving an internal hammer up and down. The hammer is first driven down to strike the chisel and then back up to return the hammer to the original position to repeat the cycle. The effectiveness of the jackhammer is dependent on how much force is applied to the tool.

The jackhammer takes it pressurized air (power source) from a compressor, the compressor takes air from the atmosphere and compresses it to a highly compressed air and force it through the hose to the jackhammer.

Jackhammer perform a simple, almost primal feat of pounding and breaking, the inner workings of jackhammers are complex. The body of a jackhammer is a vertical cylinder, which acts as a pressure chamber. Compressed air enters the chamber and activates a trigger valve, which rapidly opens and closes. When opened, the piston allows pressurized air into the piston chamber. Pressure rises in the piston chamber, causing the piston to move, striking the bit, that is the chisel.

After the piston slams into the bit, the trigger valve closes and redirects the compressed air to the cylinder below the piston. The energy of the air, along with a spring, forces the bit back into its original position. Then the process begin again.